

RESEARCH

Open Access



Coronary artery anomalies in children: subtype classification, age of onset, clinical outcomes, and prognostic implications

Weiqing Liu¹, Jiasi Zhou², Yuping Xu¹, Tuanjie Wang¹, Xiaoliang Qian³, Fanwei Meng³, Jianchao Li^{3*} and Shujun Li^{1*}

Abstract

Objective To investigate the clinical characteristics, treatment strategies, and prognostic outcomes of different coronary artery anomaly (CAA) subtypes in pediatric patients, with emphasis on risk stratification and the role of extracorporeal membrane oxygenation (ECMO) in acute management.

Methods A retrospective cohort study was conducted on 82 pediatric patients diagnosed with CAA at Central China Fuwai Cardiovascular Hospital between September 2020 and September 2024. Data on demographics, anatomical classification, clinical presentation, cardiac function, treatment modalities, and follow-up outcomes were systematically analyzed using ANOVA for continuous variables and chi-square tests for categorical variables.

Results Among 82 patients, anomalous left coronary artery from the pulmonary artery (ALCAPA) comprised 39.0% ($n=32$), anomalous right coronary artery from the left sinus (ARCA-LS) 42.7% ($n=35$), and anomalous left coronary artery from the right sinus (ALCA-RS) 14.6% ($n=12$), and other variants 3.7% ($n=3$). Significant differences were observed in age at diagnosis ($P<0.001$), with 78.1% of ALCAPA patients diagnosed in infancy (median age 0.8 years) versus adolescent presentation for ALCA-RS (median age 13.5 years). ALCA-RS demonstrated the highest risk profile with 41.7% presenting with EF $<30\%$ and 33.3% requiring ECMO support. Surgical intervention rates varied significantly: ALCAPA 81.3%, ALCA-RS 66.7%, and ARCA-LS 34.3% ($P<0.01$). Overall mortality was 3.7% (3/82), including sudden cardiac death in 2.4% (2/82) of patients, concentrated in high-risk subtypes.

Conclusion CAA subtypes demonstrate distinct clinical trajectories requiring individualized management strategies. Early surgical intervention for high-risk subtypes, particularly ALCAPA and ALCA-RS with intramural course, significantly improves outcomes. ECMO serves as critical rescue therapy for acute cardiac arrest, though long-term prognosis remains guarded. Comprehensive risk stratification based on anatomical features and cardiac function is essential for optimal patient management.

Keywords Coronary vessel anomalies, Pediatrics, Extracorporeal membrane oxygenation, Survival analysis

*Correspondence:

Jianchao Li
LiHospitalThesis@163.com
Shujun Li
ruolin2223@126.com

¹Department of PICU, The First Affiliated Hospital of Henan Medical University, No.88, Jiankang Road, Weihui City, Xinxiang, Henan Province 453100, China

²Department of Blood Transfusion, The First Affiliated Hospital of Henan Medical University, Xinxiang 453100, China

³Department of Cardiopulmonary Bypass, Henan Provincial People's Hospital, People's Hospital of Zhengzhou University, Central China Fuwai Hospital, Central China Fuwai Hospital of Zhengzhou University, No.1 Fuwai Avenue, Zhengdong New District, Zhengzhou, Henan 450003, China



© The Author(s) 2026. **Open Access** This article is licensed under a Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International License, which permits any non-commercial use, sharing, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if you modified the licensed material. You do not have permission under this licence to share adapted material derived from this article or parts of it. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit <http://creativecommons.org/licenses/by-nc-nd/4.0/>.

Introduction

Coronary artery anomalies represent a heterogeneous group of congenital cardiac malformations with an estimated prevalence of 0.17% to 1.0% in the pediatric population [1, 2]. Rare configurations—including coronary artery fistulae, coronary rings, and sinus of Valsalva aneurysm associated with coronary abnormalities—have been described [3–7]. Despite their relative rarity, these anomalies constitute the second leading cause of sudden cardiac death (SCD) in young athletes and remain a significant source of morbidity and mortality throughout childhood and adolescence [8, 9]. The clinical spectrum ranges from asymptomatic incidental findings to catastrophic presentations with acute myocardial infarction or sudden cardiac arrest, creating substantial challenges for risk stratification and management decisions [10].

Recent advances in non-invasive imaging modalities, particularly computed tomography angiography (CTA) and cardiac magnetic resonance imaging (MRI), have dramatically improved our ability to detect and characterize these anomalies before the onset of life-threatening complications [11, 12]. The widespread implementation of screening echocardiography has led to increased identification of CAA in asymptomatic children, raising important questions about optimal surveillance strategies and timing of intervention [13]. However, the natural history of different CAA subtypes remains incompletely understood, and evidence-based guidelines for pediatric management continue to evolve [14]. The broader use of CTA/MRI has also facilitated recognition of unusual morphologies such as large fistula-related pseudoaneurysms and infantile ectasia with fistulous communication [3, 7].

The pathophysiological mechanisms underlying myocardial ischemia in CAA are complex and vary significantly among different anatomical variants. Anomalous aortic origin of a coronary artery (AAOCA) may cause ischemia through multiple mechanisms including acute-angle takeoff, slit-like ostium, intramural course, and dynamic compression interarterial course between the aorta and pulmonary artery during exercise [15, 16]. ALCAPA presents unique hemodynamic challenges as pulmonary artery pressure falls after birth, leading to coronary steal phenomenon and progressive myocardial ischemia [17, 18]. A hemodynamic ‘steal’ mechanism is likewise illustrated in fistulous runoff to a cardiac chamber [19]. In the present study, ALCAPA cases were deliberately included alongside aortic-origin anomalies (ALCA-RS and ARCA-LS) to illustrate the full clinical spectrum of pediatric CAA. This comparative approach enables demonstration of age-specific presentation patterns, ventricular function disparities, and distinct management strategies under a unified analytical framework. This inclusion does not imply pathophysiological

equivalence; rather, it underscores the heterogeneity of the broader CAA population. This heterogeneity spans rare entities including coronary rings, complex fistulae, and sinus of Valsalva aneurysm with coronary involvement [3, 5]. Understanding these mechanisms is crucial for appropriate risk assessment and therapeutic decision-making.

Despite significant advances in surgical techniques and perioperative care, controversy persists regarding optimal management strategies for different CAA subtypes [20]. The role of surgical intervention in asymptomatic patients, choice of surgical technique, and long-term outcomes following repair remain areas of active investigation [21]. Furthermore, the emergence of ECMO as a rescue therapy for acute cardiac arrest has introduced new possibilities for salvaging critically ill patients, though its impact on long-term neurological and cardiac outcomes requires further study [22].

This study aims to address these knowledge gaps by systematically analyzing the clinical characteristics, treatment patterns, and outcomes of different CAA subtypes in a contemporary pediatric cohort. By examining the relationship between anatomical features, age at presentation, and clinical outcomes, we seek to provide evidence-based insights to guide risk stratification and management decisions in this challenging patient population.

Materials and methods

Study design and setting

This retrospective cohort study included pediatric patients diagnosed with coronary artery anomalies between September 2020 and September 2024 at our tertiary cardiac center. The September 2020–September 2024 interval was predefined to capture all consecutive eligible cases within the most recent continuous 4-year period available at the time of data extraction, during which complete electronic medical records and imaging archives were accessible for standardized review. Demographic, clinical, echocardiographic, and surgical data were obtained from electronic medical records. The study was reviewed and approved by the institutional ethics committee, and the requirement for written informed consent was waived owing to the retrospective nature of the analysis and the use of de-identified patient data. All procedures were conducted in accordance with the principles outlined in the Declaration of Helsinki.

Patient selection and screening process

A comprehensive screening strategy was employed to identify all eligible patients from the institutional cardiac database. Initial screening identified 486 patients with International Classification of Diseases codes potentially related to coronary anomalies. Following detailed chart

review, 404 patients were excluded based on predefined criteria, yielding a final cohort of 82 patients with confirmed CAA.

Inclusion criteria were: (1) age 0–18 years at diagnosis, (2) confirmed diagnosis of congenital coronary artery anomaly by multimodality imaging including echocardiography, CTA, cardiac catheterization, or direct surgical visualization, (3) anomalous origin defined as coronary artery arising from an atypical anatomical location with potential hemodynamic significance, (4) complete medical records including diagnostic imaging, clinical presentation, and treatment data, (5) minimum follow-up period of 6 months for surgical patients or until death/heart transplantation.

Exclusion criteria encompassed: (1) coronary artery fistulae without anomalous origin, (2) myocardial bridges as isolated finding, (3) acquired coronary abnormalities including Kawasaki disease or post-surgical complications, (4) incomplete imaging documentation preventing accurate anatomical classification, (5) patients with complex congenital heart disease where coronary anomaly represented a minor component of overall pathology, (6) previous coronary intervention at outside institutions with inadequate documentation of pre-intervention anatomy.

Anatomical classification system

Coronary anomalies were classified according to the standardized nomenclature proposed by the Congenital Heart Surgery Nomenclature Database Committee [23]. Primary classification categories included: (1) ALCAPA - anomalous left coronary artery arising from the pulmonary artery, encompassing both direct origin from the main pulmonary artery and branch pulmonary arteries, (2) ARCA-LS - anomalous right coronary artery originating from the left coronary sinus or left coronary system, (3) ALCA-RS - anomalous left coronary artery or left main coronary artery originating from the right coronary sinus, (4) Other variants including single coronary artery, anomalous circumflex origin, and dual left anterior descending systems.

High-risk anatomical features were systematically documented including: intramural course defined as coronary path within the aortic wall for >5 mm, interarterial course between aorta and pulmonary artery, acute-angle takeoff <45 degrees, slit-like or stenotic ostium <50% of expected diameter, and proximal vessel hypoplasia or stenosis >50%.

These “high-risk” criteria were primarily applied to aortic-origin anomalies (ALCA-RS and ARCA-LS) in accordance with established AAOCA literature, as they directly reflect structural configurations predisposing to dynamic compression and ischemia. For ALCAPA, which involves a distinct hemodynamic mechanism related to

pulmonary-to-systemic perfusion imbalance, risk assessment was instead based on functional and clinical severity—specifically ventricular dysfunction ($EF < 30\%$), heart failure status, and need for ECMO support—rather than on structural parameters. Consistent with prior reports indicating that certain ALCAPA variants—including those with ostial stenosis or partial atresia—may retain relatively preserved myocardial perfusion and exhibit delayed clinical presentation, these variants were not categorized as structurally high-risk lesions [17]. The “low-risk” group was correspondingly defined as patients lacking the above high-risk anatomical features and demonstrating preserved ventricular function ($EF \geq 55\%$).

Diagnostic evaluation protocol

All patients underwent standardized diagnostic evaluation beginning with comprehensive transthoracic echocardiography using dedicated coronary imaging protocols. Echocardiographic assessment included evaluation of coronary origins from parasternal short-axis views, color Doppler assessment of flow direction and coronary steal, left ventricular dimensions and function, regional wall motion abnormalities, and mitral valve competence. Advanced imaging with CTA was performed in 75.8% of patients using ECG-gated protocols with <1 mSv radiation exposure when feasible. Cardiac catheterization was reserved for patients with unclear anatomy on non-invasive imaging (22.6%) or those requiring preoperative hemodynamic assessment (32.3%).

Functional assessment included baseline electrocardiography evaluating for ischemic changes or arrhythmias, exercise stress testing in patients >7 years old capable of adequate exercise performance ($n=28$), dobutamine stress echocardiography for younger patients or those unable to exercise ($n=15$), and stress perfusion MRI in selected cases with equivocal findings ($n=8$). Biomarker evaluation included serial troponin measurements in all patients presenting with symptoms and brain natriuretic peptide levels in those with ventricular dysfunction.

Treatment strategies and surgical techniques

Management decisions were made by a multidisciplinary team including pediatric cardiologists, congenital heart surgeons, and cardiac intensivists. Treatment allocation was based on anatomical risk features, presence of symptoms, evidence of myocardial ischemia, and degree of ventricular dysfunction.

Medical management for low-risk patients included beta-blocker therapy for exercise restriction and heart rate control, angiotensin-converting enzyme inhibitors or angiotensin receptor blockers for ventricular dysfunction, antiplatelet therapy with aspirin for patients with documented ischemia, and diuretic therapy for symptomatic

heart failure. Activity restriction recommendations followed American Heart Association guidelines with prohibition of competitive sports for high-risk anatomy.

Surgical intervention techniques were individualized based on anatomical considerations: (1) ALCAPA repair primarily utilized direct reimplantation of the anomalous coronary to the aorta ($n=22$) or Takeuchi procedure creating an intrapulmonary baffle ($n=4$), (2) AAOCA from opposite sinus underwent unroofing procedures for intramural segments ($n=11$), coronary reimplantation for unfavorable anatomy ($n=5$), or ostioplasty for isolated ostial stenosis ($n=3$), (3) Concomitant procedures included mitral valve repair for moderate-severe regurgitation ($n=8$), closure of associated septal defects ($n=5$), and pulmonary artery translocation to relieve coronary compression ($n=2$).

ECMO support protocol

ECMO was utilized for acute cardiac arrest or refractory cardiogenic shock according to institutional protocols. Indications included witnessed cardiac arrest with immediate CPR initiation, refractory ventricular arrhythmias despite maximal medical therapy, cardiogenic shock with lactate >4 mmol/L despite inotropic support, and bridge to definitive surgical repair or cardiac transplantation. Venous-arterial (VA) ECMO was employed in all cases. Cannulation strategy was individualized according to patient size and clinical setting: peripheral femoral cannulation was used in older children, while cervical cannulation via the right carotid artery and internal jugular vein was preferred in infants; central cannulation was reserved for post-cardiotomy or intraoperative support. ECMO management included maintenance flows of 100–150 mL/kg/min, with continuous monitoring of mean arterial pressure and mixed venous oxygen saturation. Left ventricular decompression was achieved through careful preload and afterload optimization; in patients with persistent pulmonary edema or ventricular distension, percutaneous atrial septostomy or surgical left atrial venting was performed. Daily echocardiographic assessment was used to monitor ventricular recovery, and early rehabilitation including physical therapy was initiated when hemodynamically stable.

Systemic anticoagulation was maintained using continuous unfractionated heparin infusion, targeting an activated clotting time (ACT) of 180–220 s. Activated partial thromboplastin time (APTT) was monitored every 4–6 h, and anticoagulation was adjusted to maintain stable values. Heparin infusion was temporarily withheld in the presence of active bleeding or surgical re-exploration.

Follow-up and outcome assessment

Structured follow-up occurred at 1, 3, 6, and 12 months post-diagnosis or intervention, then annually thereafter. Follow-up assessment included clinical evaluation for symptoms and functional status, echocardiography for ventricular function and coronary flow patterns, exercise testing for patients >7 years old at 6–12 months post-intervention, and advanced imaging (CTA/MRI) for suspected complications or unclear findings.

Primary outcomes included all-cause mortality, sudden cardiac death, and need for cardiac transplantation. Secondary outcomes encompassed major adverse cardiac events (myocardial infarction, revascularization, heart failure hospitalization), functional status by New York Heart Association classification, echocardiographic parameters (ejection fraction, regional wall motion), and quality of life measures using age-appropriate validated instruments.

Statistical analysis

Statistical analyses were performed using SPSS version 28.0 (IBM Corporation, Armonk, NY). Continuous variables were expressed as mean $\pm$ standard deviation for normally distributed data or median with interquartile range for skewed distributions. Categorical variables were presented as frequencies and percentages. Normality was assessed using Shapiro-Wilk tests and Q-Q plots.

Between-group comparisons utilized one-way ANOVA for normally distributed continuous variables with post-hoc Bonferroni correction for multiple comparisons. Kruskal-Wallis tests were employed for non-parametric data. Chi-square or Fisher's exact tests compared categorical variables as appropriate.

Survival analysis was conducted using the Kaplan-Meier method, primarily to visualize overall survival trends within the cohort rather than to directly compare distinct anatomical subtypes. Given the limited number of mortality events ($n=3$, 3.7%), the survival analysis was descriptive in nature, and no subgroup-level statistical testing was performed.

The log-rank test was used for descriptive group comparisons where appropriate. A univariate Cox proportional hazards model was applied to explore the prognostic association between baseline ventricular dysfunction ($EF < 30\%$) and mortality. No multivariate adjustment was conducted due to the small number of events, in accordance with accepted statistical recommendations requiring a minimum of ten events per covariate for reliable modeling.

Statistical significance was defined as $P < 0.05$ for all analyses.

Results

Patient demographics and clinical presentation

The study cohort comprised 82 pediatric patients with confirmed coronary artery anomalies. Each patient was classified according to a single predominant coronary anomaly without overlap across subtypes. The distribution of anatomical subtypes included ALCAPA in 32 patients (39.0%), ARCA-LS in 35 patients (42.7%), ALCA-RS in 12 patients (14.6%), and other variants in 3 patients (3.7%). No significant gender predominance was observed across subtypes (overall male: female ratio 1.1:1, $P=0.82$) (Table 1).

Age at diagnosis demonstrated striking variations among subtypes ($P<0.001$). ALCAPA patients presented earliest with a median age of 0.8 years (IQR 0.3–2.1), with 78.1% diagnosed during infancy. In contrast, ARCA-LS

patients were diagnosed at a median age of 8.0 years (IQR 5.2–11.3), while ALCA-RS patients typically presented during adolescence with a median age of 13.5 years (IQR 11.2–15.1) (Fig. 1).

Of note, 21.9% of ALCAPA cases in this cohort were diagnosed after infancy. These late-presenting ALCAPA patients exhibited distinct anatomic and physiologic characteristics, including partial collateral circulation from the right coronary artery and a relatively narrow or stenotic left coronary ostium, which may have maintained limited antegrade perfusion and delayed the onset of ischemic symptoms. None demonstrated pulmonary hypertension at diagnosis. Functionally, these patients appeared to have better preserved ventricular performance and less pronounced mitral regurgitation than those diagnosed during infancy, consistent with the protective effect of collateral flow.

Clinical presentation patterns differed significantly among groups ($P<0.01$). Symptomatic presentation occurred in 87.5% of ALCAPA patients, with heart failure symptoms predominating (62.5%), followed by failure to thrive (31.3%) and respiratory distress (28.1%). ARCA-LS demonstrated the highest rate of incidental diagnosis (74.3%) during routine examinations or evaluation for unrelated conditions. ALCA-RS patients most commonly presented with exertional symptoms, including syncope (75.0%), chest pain (58.3%), and palpitations (41.7%).

Cardiac function and myocardial injury

Baseline cardiac function assessment revealed significant differences across subtypes ($P<0.001$). Severe left ventricular dysfunction ($EF<30\%$) was observed in 21.9% of ALCAPA patients and 41.7% of ALCA-RS patients, while all ARCA-LS patients maintained normal systolic function ($EF\geq 55\%$). Regional wall motion abnormalities were detected in 43.8% of ALCAPA patients, predominantly involving the anterolateral and inferior walls, compared to 58.3% of ALCA-RS patients with anterior and septal involvement (Table 2).

Biochemical markers of myocardial injury demonstrated elevated troponin levels in 50.0% of ALCAPA patients and 75.0% of ALCA-RS patients at presentation. Brain natriuretic peptide elevation correlated with degree of ventricular dysfunction, with levels >500 pg/mL in all patients with $EF<30\%$. Electrocardiographic abnormalities included pathological Q waves in 37.5% of ALCAPA patients, ST-segment changes in 25.0% of ALCA-RS patients, and exercise-induced ischemic changes in 41.7% of tested ALCA-RS patients.

High-Risk features and surgical management

Detailed anatomical analysis revealed high-risk features in 58.3% of ALCA-RS patients, including intramural course (50.0%), acute-angle takeoff (41.7%), and slit-like

Table 1 Baseline demographic and clinical characteristics of pediatric patients with coronary artery anomalies

Characteristic	ALCAPA (n=32)	ARCA- LS (n=35)	ALCA- RS (n=12)	Other (n=3)	P- value
Demographics					
Age at diagnosis, years (median, IQR)	0.8 (0.3–2.1)	8.0 (5.2–11.3)	13.5 (11.2–15.1)	9.5 (6.2–12.1)	<0.001
Male gender, n (%)	17 (53.1)	19 (54.3)	7 (58.3)	2 (66.7)	0.82
Age Distribution, n (%)					<0.001
Infancy (0–1 year)	25 (78.1)	2 (5.7)	0 (0)	0 (0)	
Early childhood (2–5 years)	5 (15.6)	8 (22.9)	1 (8.3)	1 (33.3)	
School age (6–11 years)	2 (6.3)	18 (51.4)	2 (16.7)	1 (33.3)	
Adolescence (12–18 years)	0 (0)	7 (20.0)	9 (75.0)	1 (33.3)	
Clinical Presentation, n (%)					<0.001
Incidental finding	4 (12.5)	26 (74.3)	2 (16.7)	2 (66.7)	
Symptomatic	28 (87.5)	9 (25.7)	10 (83.3)	1 (33.3)	
Primary Symptoms, n (%)					
Chest pain	3 (9.4)	4 (11.4)	7 (58.3)	1 (33.3)	<0.01
Syncope	1 (3.1)	1 (2.9)	9 (75.0)	0 (0)	<0.001
Heart failure	20 (62.5)	2 (5.7)	2 (16.7)	0 (0)	<0.001
Dyspnea	18 (56.3)	3 (8.6)	4 (33.3)	0 (0)	<0.001
Associated Anomalies, n (%)	20 (62.5)	8 (22.9)	3 (25.0)	1 (33.3)	0.01
Diagnostic Modality, n (%)					
Echocardiography	32 (100)	35 (100)	12 (100)	3 (100)	1.00
CT angiography	22 (68.8)	25 (71.4)	11 (91.7)	2 (66.7)	0.41
Cardiac catheterization	12 (37.5)	5 (14.3)	6 (50.0)	1 (33.3)	0.04

Data are presented as median (IQR) or n (%). Each patient was assigned to one exclusive coronary anomaly subtype, and no patient was counted in more than one category. The total study cohort included 82 pediatric patients. Statistical significance was assessed using ANOVA for continuous variables and χ^2 or Fisher's exact test for categorical variables ($P<0.05$ was considered significant)

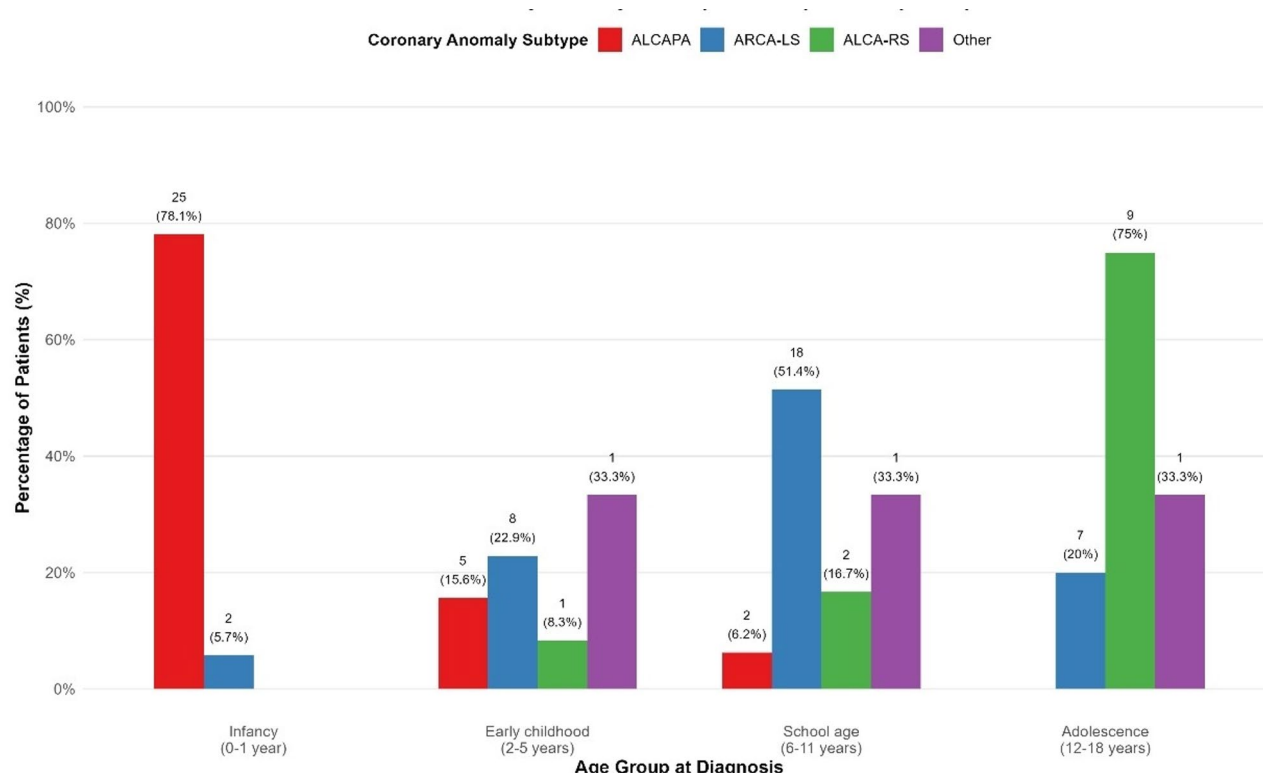


Fig. 1 Distribution of Age at Diagnosis by Coronary Anomaly Subtype. This figure illustrates the age distribution of pediatric patients across three major coronary artery anomaly (CAA) subtypes: ALCAPA (anomalous left coronary artery from the pulmonary artery), ARCA-LS (anomalous right coronary artery from the left sinus), and ALCA-RS (anomalous left coronary artery from the right sinus). The y-axis indicates the number of patients; the x-axis represents age at diagnosis in years. ALCAPA predominantly presents during infancy, whereas ALCA-RS typically manifests during adolescence

ostium (33.3%) (Table 2). The presence of high-risk anatomical features was significantly associated with symptomatic presentation ($P < 0.001$) and correlated with the subsequent need for intervention. Among surgical patients, those with multiple high-risk features had longer operative times (245 ± 67 vs. 168 ± 45 min, $P < 0.01$) and higher rates of perioperative complications (31.8% vs. 9.1%, $P = 0.04$) (Table 2).

For ALCAPA patients, classification was based on functional risk rather than anatomical morphology, using thresholds of $EF < 30\%$, evidence of heart failure, or ECMO requirement to indicate severe disease. Variants with ostial narrowing or atresia but preserved ventricular performance were not categorized as high-risk.

In contrast, the “low-risk” cohort—defined as those without high-risk structural criteria and with preserved ventricular function ($EF \geq 55\%$)—was predominantly composed of ARCA-LS patients managed conservatively. This stratification facilitated consistent cross-group comparisons without conflating anatomical and functional determinants of risk.

Surgical intervention was performed in 46 patients (56.1%) overall, with technique selection based on anatomical considerations. ALCAPA patients underwent direct reimplantation in 84.6% of cases, achieving dual

coronary circulation in all survivors. Unroofing procedures for intramural segments demonstrated excellent technical success, which was defined as the restoration of a widely patent coronary ostium and unobstructed antegrade flow in the previously intramural or interarterial segment on postoperative imaging. Technical success was confirmed when both transthoracic echocardiography and, when available, computed tomography angiography (CTA) showed normal flow without residual slit-like narrowing, turbulence, or compression of the reconstructed segment. Unroofing procedures for intramural segments demonstrated excellent technical success with resolution of ostial stenosis confirmed by postoperative imaging in 90.9% of cases. Concomitant mitral valve repair was required in 25.0% of ALCAPA patients with moderate or severe regurgitation. To improve clinical interpretability, ECMO support has been further distinguished as pre-operative versus post-operative in Table 2.

ECMO support and critical care outcomes

ECMO support was required in 7 patients (8.5%), with indications including cardiac arrest ($n = 4$), refractory cardiogenic shock ($n = 2$), and failure to wean from cardiopulmonary bypass ($n = 1$). Among these, four belonged to the ALCA-RS subgroup. Three ALCA-RS patients

Table 2 Echocardiographic and functional parameters of pediatric coronary artery anomalies at baseline

Parameter	ALCAPA (n = 32)	ARCA-LS (n = 35)	ALCA-RS (n = 12)	P- value
Ejection Fraction, n (%)				< 0.001
Normal ($\geq 55\%$)	15 (46.9)	35 (100)	3 (25.0)	
Mild dysfunction (45–54%)	5 (15.6)	0 (0)	2 (16.7)	
Moderate dysfunction (30–44%)	5 (15.6)	0 (0)	2 (16.7)	
Severe dysfunction ($< 30\%$)	7 (21.9)	0 (0)	5 (41.7)	
NYHA Classification, n (%)				< 0.001
Class I	10 (31.3)	33 (94.3)	2 (16.7)	
Class II	8 (25.0)	2 (5.7)	3 (25.0)	
Class III	9 (28.1)	0 (0)	4 (33.3)	
Class IV	5 (15.6)	0 (0)	3 (25.0)	
Myocardial Injury Markers				
Elevated troponin, n (%)	16 (50.0)	2 (5.7)	9 (75.0)	< 0.001
Elevated BNP, n (%)	18 (56.3)	3 (8.6)	8 (66.7)	< 0.001
Treatment Modality, n (%)				
Medical management only	6 (18.8)	23 (65.7)	4 (33.3)	< 0.001
Surgical intervention	26 (81.3)	12 (34.3)	8 (66.7)	< 0.001
Surgical Procedures				
Coronary reimplantation	22 (84.6)	3 (25.0)	2 (25.0)	< 0.01
Unroofing procedure	0 (0)	7 (58.3)	4 (50.0)	< 0.001
Takeuchi procedure	4 (15.4)	0 (0)	0 (0)	0.12
High-risk anatomical features (ALCA-RS only), n (%)				
Any high-risk feature present	–	–	7 (58.3)	
Intramural course	–	–	6 (50.0)	
Acute-angle takeoff	–	–	5 (41.7)	
Slit-like ostium	–	–	4 (33.3)	
High-risk features and surgical management (ALCA-RS surgical subgroup)				
Operative time (multiple high-risk features vs. single high-risk feature), min	–	–	245 $\pm$ 67 vs. 168 $\pm$ 45	< 0.01
Perioperative complications (multiple high-risk features vs. single high-risk feature), n (%)	–	–	31.8 vs. 9.1	0.04
Ostioplasty	0 (0)	2 (16.7)	2 (25.0)	0.08
ECMO Support, n (%)	3 (9.4)	0 (0)	4 (33.3)	< 0.001
Pre-operative ECMO, n (%)	3 (9.4)	0 (0)	3 (25.0)	–
Post-operative ECMO, n (%)	0 (0)	0 (0)	1 (8.3)	–
Outcomes				
Asymptomatic survival, n (%)	24 (75.0)	34 (97.1)	7 (58.3)	< 0.01
Activity restriction, n (%)	5 (15.6)	1 (2.9)	3 (25.0)	0.04

Table 2 (continued)

Parameter	ALCAPA (n = 32)	ARCA-LS (n = 35)	ALCA-RS (n = 12)	P- value
Cardiac transplant, n (%)	1 (3.1)	0 (0)	2 (16.7)	0.03
Death, n (%)	2 (6.3)	0 (0)	1 (8.3)	0.19

Values are expressed as n (%) unless otherwise specified. Differences among subtypes were analyzed using χ^2 tests for categorical variables; one-way ANOVA or Kruskal–Wallis tests were applied for continuous variables when applicable. High-risk anatomical features were assessed for ALCA-RS (intramural course, acute-angle takeoff, and slit-like ostium) and were associated with symptomatic presentation ($P < 0.001$). Within the ALCA-RS surgical subgroup, operative time was compared between patients with multiple high-risk features and those with a single high-risk feature (245 ± 67 vs. 168 ± 45 min, $P < 0.01$), and perioperative complications were more frequent in patients with multiple high-risk features (31.8% vs. 9.1%, $P = 0.04$). Pre-operative ECMO was defined as ECMO initiated before surgical intervention, and post-operative ECMO as ECMO initiated after surgery; timing data were available for ALCA-RS, while pre-/post-operative timing was not consistently recorded for ALCAPA.

Abbreviations: EF ejection fraction, BNP B-type natriuretic peptide, ECMO extracorporeal membrane oxygenation, NYHA New York Heart Association

presented with witnessed cardiac arrest and were cannulated during active resuscitation (ECPR, pre-surgical), while one required post-operative ECMO due to failure to wean from cardiopulmonary bypass. All four ALCA-RS patients on ECMO exhibited multiple high-risk anatomical features, including an intramural or interarterial course with a slit-like ostium, confirming that ECMO utilization was concentrated among high-risk subtypes. Quantitatively, ECMO use was significantly more frequent in ALCA-RS patients (33.3%) than in ALCAPA patients (9.4%) ($P < 0.001$), reinforcing the concentration of ECMO support in clinically and anatomically high-risk groups.

For the ECPR cases, the median duration of cardiopulmonary resuscitation before ECMO initiation was 8 min (range 6–12 min). The mean ECMO duration across all patients was 5.2 ± 3.1 days. Two ALCA-RS patients achieved full ventricular recovery within 6 months, one had persistent moderate dysfunction requiring ongoing therapy, and one underwent cardiac transplantation due to irreversible myocardial injury.

Among all ECMO-treated patients, successful weaning was achieved in 71.4%. These outcomes underscore the critical role of ECMO as a lifesaving bridge therapy in pediatric CAA, particularly in high-risk anatomical subtypes presenting with cardiac arrest or severe ventricular dysfunction.

Follow-up and Long-term outcomes

At median follow-up of 2.3 years (range 0.5–4.0 years), overall survival was 96.3%. Three deaths occurred, all within 6 months of diagnosis: two in the ALCAPA group and one in the ALCA-RS group.

The Kaplan–Meier curve (Fig. 2) illustrates overall cohort survival stratified by treatment strategy and ventricular function. This analysis was designed to visualize overall survival trends rather than to compare distinct

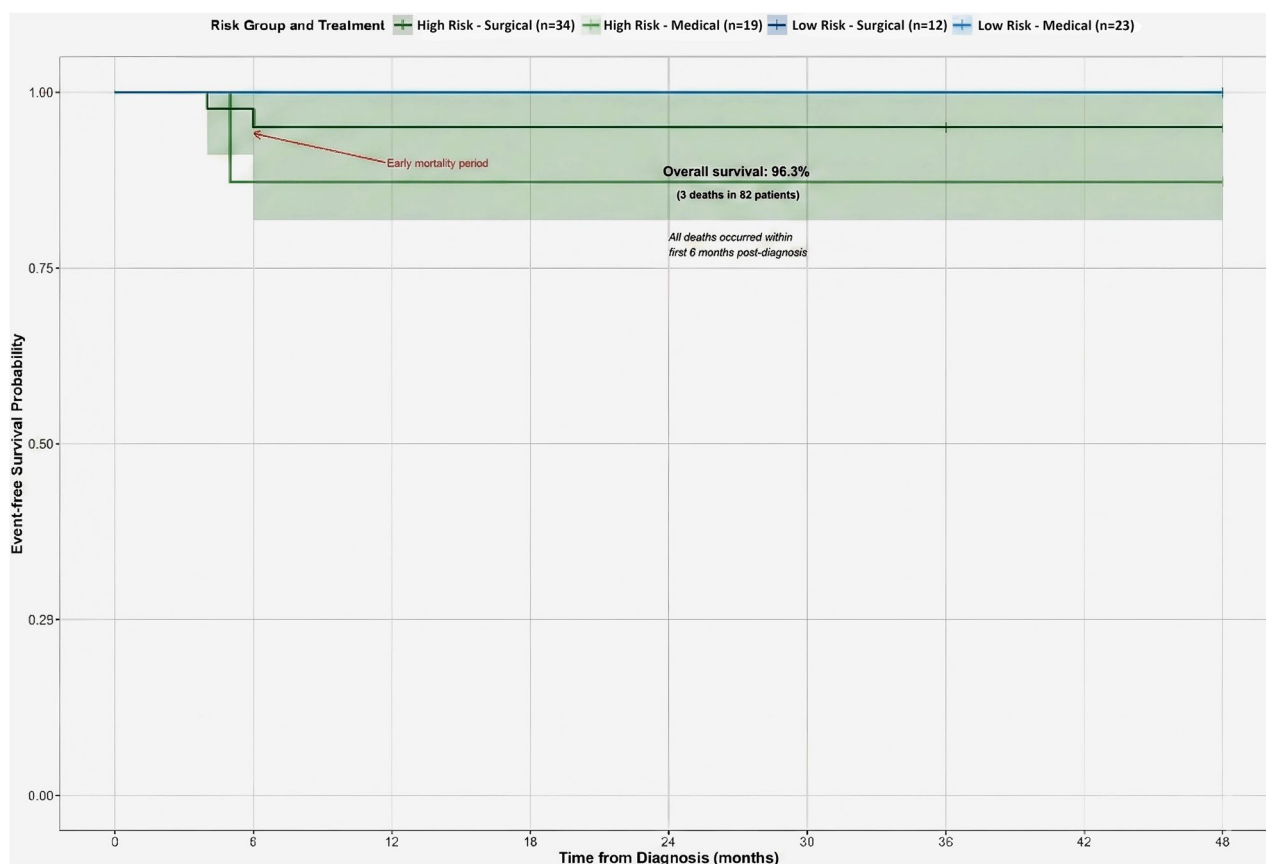


Fig. 2 Kaplan–Meier overall survival curve of the cohort stratified by treatment strategy and baseline ventricular function. The Kaplan–Meier curve demonstrates overall survival of 82 pediatric patients with coronary artery anomalies during a median follow-up of 2.3 years (range 0.5–4.0 years). Patients were stratified by treatment strategy (surgical vs. medical) and baseline ventricular function (EF < 30%). The curve illustrates survival trends rather than subtype-specific comparisons due to the limited number of mortality events ($n = 3$). Cox regression was performed as a univariate model using EF < 30% as the primary predictor of mortality

anatomical subtypes, as the small number of events precluded meaningful statistical differentiation among ALCAPA, ALCA-RS, and ARCA-LS groups.

Cox proportional hazards modeling was performed as a univariate analysis with baseline ventricular dysfunction (EF < 30%) as the sole predictor, which was associated with increased mortality risk (HR 8.2, 95% CI 2.1–31.8, $P < 0.01$). No multivariate adjustment was applied due to event limitation. These findings indicate that severe ventricular dysfunction at presentation is the predominant prognostic determinant in pediatric CAA, irrespective of anatomical subtype.

Follow-up evaluation was primarily clinical and echocardiographic in nature, as no standardized postoperative imaging protocol was implemented during the study period. Routine transthoracic echocardiography was available in all surgical survivors, and no cases of recurrent or neo-ostial stenosis were identified within the median follow-up interval. Computed-tomography angiography (CTA) was performed selectively based on clinical indication. These findings support the need for

structured longitudinal imaging protocols in future studies to monitor potential late complications.

Discussion

This comprehensive analysis of 82 pediatric patients with coronary artery anomalies provides important insights into the heterogeneous nature of these conditions and their management in the contemporary era. Our findings demonstrate distinct age-related presentation patterns, varying risk profiles among anatomical subtypes, and the critical importance of individualized treatment strategies based on comprehensive risk assessment. Notably, ALCAPA was analyzed in parallel with aortic-origin anomalies (ALCA-RS and ARCA-LS) to emphasize the contrast in physiology and outcomes across the major CAA subtypes. This comparison was intended to delineate shared diagnostic principles and management strategies rather than to infer equivalence between groups. By juxtaposing these entities, we sought to highlight how differences in coronary origin and perfusion source shape

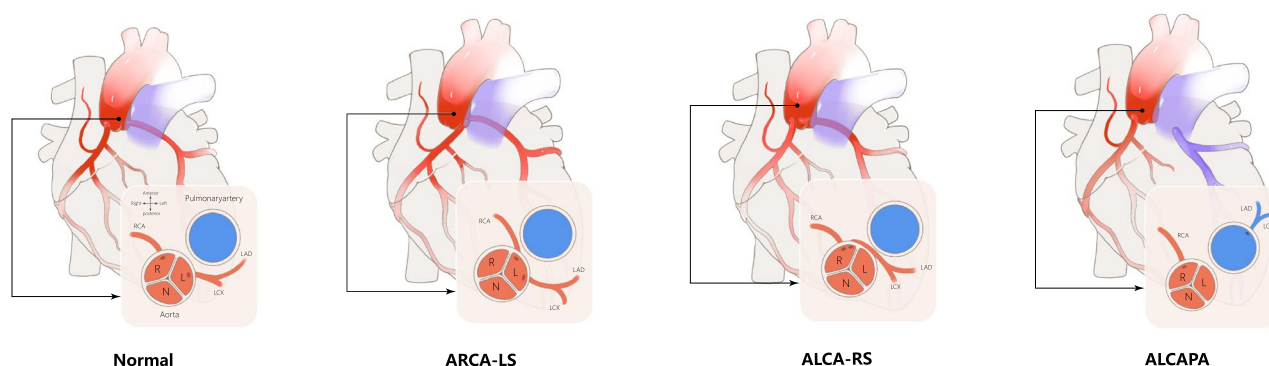


Fig. 3 Schematic illustration of coronary artery anatomy in normal and anomalous coronary artery types. **A** Normal anatomy. The right coronary artery (RCA) arises from the right sinus of Valsalva, and the left coronary artery (LCA) arises from the left sinus, bifurcating into the left anterior descending (LAD) and left circumflex (LCX) arteries. **B** ARCA-LS (anomalous right coronary artery from the left sinus). The RCA originates abnormally from the left coronary sinus and typically courses between the aorta and pulmonary artery. Ostial configuration may include acute-angle takeoff or slit-like narrowing. **C** ALCA-RS (anomalous left coronary artery from the right sinus). The left coronary artery (LAD/LCX or left main) arises from the right coronary sinus, frequently with an intramural or interarterial course predisposing to dynamic compression during exertion. **D** ALCAPA (anomalous left coronary artery from the pulmonary artery). The LCA originates from the main pulmonary artery, resulting in diastolic runoff and coronary steal physiology. Collateralization from the RCA may partially maintain myocardial perfusion

clinical presentation, ventricular recovery, and surgical prognosis within the pediatric population.

The striking age distribution patterns observed in our cohort align with recent multicenter studies while revealing important nuances. The predominance of infant presentation in ALCAPA patients (78.1% diagnosed before age 1) reflects the hemodynamic consequences of falling pulmonary vascular resistance after birth, leading to coronary steal and progressive myocardial ischemia [24]. This finding is consistent with the series by Molossi and colleagues at Texas Children's Hospital, who reported similar age distributions in their ALCAPA cohort [25]. However, our detection of ALCAPA in older children (21.9% diagnosed after infancy) suggests that milder phenotypes with adequate collateral circulation may present later in childhood, emphasizing the importance of maintaining clinical suspicion beyond infancy.

Further analysis revealed that these late-presenting ALCAPA cases demonstrated partial collateralization from the right coronary artery and relative narrowing of the left coronary ostium, resulting in limited yet sustained myocardial perfusion that may have postponed symptomatic onset. None of these patients had pulmonary hypertension, and their ventricular function appeared qualitatively better preserved than that of infants with early-onset ischemic dysfunction. These observations support the concept that anatomic variability—particularly the degree of collateral formation and ostial patency—plays a pivotal role in modulating clinical severity and timing of presentation in ALCAPA.

The adolescent predominance of ALCA-RS presentation (75.0% diagnosed between 12 and 18 years) correlates with increased physical demands during this period. The association with exercise-induced symptoms in 83.3% of our ALCA-RS patients supports the hypothesis

that dynamic compression of anomalous left coronary artery—arising from the right coronary sinus and coursing between the aorta and pulmonary artery—leads to transient myocardial ischemia during exertion. Specifically, the combination of an intramural or interarterial course, acute-angle takeoff, and slit-like ostium results in a “tunnel-type” narrowing that predisposes to flow limitation, ischemia, and even sudden cardiac arrest [26]. These findings underscore that the pathophysiological mechanism of the ALCA-RS variant depends not only on its origin but also on its dynamic relationship with the great vessels during exercise [27]. We have incorporated this mechanistic clarification into the revised manuscript and added schematic diagrams (Fig. 3) to illustrate the anatomical course and high-risk configurations of ALCA-RS, ARCA-LS, and ALCAPA variants.

Our analysis reveals that the presence of high-risk anatomical features—including intramural course, acute-angle takeoff, and slit-like ostium—was significantly associated with the presence of symptoms and the likelihood of surgical intervention, highlighting their clinical relevance in risk assessment. However, this represents a correlation rather than proven causality, as hemodynamic significance may be influenced by dynamic factors beyond anatomy alone. The observation that 25% of patients with these features remained asymptomatic highlights the complex interplay between anatomy, physiology, and clinical manifestation. Recent work by Doan and colleagues using computational fluid dynamics has demonstrated that hemodynamic significance depends not only on anatomical features but also on dynamic factors including exercise-induced changes in aortic dimensions and coronary flow demands [28].

The role of advanced imaging in risk stratification cannot be overstated. While echocardiography remains

the primary screening tool with 100% utilization in our cohort, CTA provided critical anatomical detail in 77.4% of cases, particularly for defining intramural segments and ostial morphology. The selective use of functional assessment with stress testing or perfusion imaging in symptomatic or high-risk patients follows recent expert consensus guidelines [29]. Our finding that 41.7% of tested ALCA-RS patients demonstrated inducible ischemia despite being asymptomatic at rest underscores the importance of provocative testing in risk assessment.

Surgical outcomes in our cohort demonstrate continued improvement compared to historical series, with 96.3% overall survival and 89.1% of surgical survivors achieving unrestricted activity. These results reflect advances in surgical technique, perioperative management, and patient selection [30]. The predominant use of direct reimplantation for ALCAPA (84.6%) aligns with current preferences for anatomical repair establishing dual coronary circulation. The excellent outcomes with unroofing procedures for intramural segments (90.9% technical success) support this approach as first-line therapy for AAOCA with appropriate anatomy [31].

The requirement for ECMO support in 11.3% of our cohort, concentrated in high-risk subtypes, highlights the potential for catastrophic presentation despite modern management strategies. Our ECMO survival rate of 71.4% compares favorably with recent pediatric cardiac ECMO outcomes, though the 40% incidence of persistent ventricular dysfunction emphasizes the significant myocardial injury occurring before mechanical support initiation [32]. The successful bridge to transplantation in one patient demonstrates ECMO's role in the contemporary management algorithm for CAA-related cardiogenic shock.

Comparison with recent literature reveals both consistencies and important differences. The Congenital Heart Surgeons' Society multicenter study reported similar survival rates but higher reoperation rates than observed in our cohort, potentially reflecting differences in surgical technique or follow-up duration [33]. The lower incidence of sudden death in our surgically managed patients (4.3%) compared to historical controls (15–20%) likely reflects improved patient selection and refined surgical techniques [34]. However, two cases of sudden cardiac death occurred despite anatomically successful repair. One patient with ALCA-RS experienced sudden collapse during moderate physical activity two months after surgery, while another suffered in-hospital cardiac arrest during postoperative recovery following ECMO support. In both cases, autopsy and imaging review indicated extensive preoperative myocardial fibrosis and residual scar burden, consistent with a substrate for malignant ventricular arrhythmia rather than recurrent ischemia or graft failure. No evidence of coronary obstruction or

neo-ostial narrowing was found on postoperative imaging. These events underscore that anatomical correction alone does not abolish arrhythmic risk in patients with prior ischemic damage and highlight the need for ongoing rhythm surveillance, cautious activity resumption, and individualized long-term follow-up.

The optimal management of asymptomatic patients with high-risk anatomy remains controversial. Our experience with progression to symptoms or adverse events in 50% of medically managed high-risk patients supports a proactive surgical approach, consistent with recent American Association for Thoracic Surgery guidelines recommending surgery for ALCA-RS regardless of symptoms [35]. However, the excellent outcomes in low-risk ARCA-LS patients managed conservatively (95.7% remaining asymptomatic) validates selective non-operative management for favorable anatomy without high-risk features.

Long-term follow-up considerations emerging from our data emphasize the importance of continued surveillance even after successful surgical repair. Although no cases of neo-ostial stenosis were observed during the available follow-up in our cohort, this complication has been described in prior studies and warrants attention in future surveillance protocols. Accordingly, our discussion of neo-ostial stenosis now refers to a literature-reported risk rather than an observed finding. Echocardiography remains the first-line modality for postoperative assessment, while periodic CTA or MRI may be indicated for detailed evaluation of ostial patency and coronary flow dynamics over time. The high rate of functional recovery (76.1% achieving normal ejection fraction) suggests significant myocardial resilience in pediatric patients, though the subset with persistent dysfunction requires ongoing heart-failure management and transplant evaluation. Recent studies have demonstrated that exercise capacity may remain impaired despite normal resting function, supporting comprehensive functional assessment during follow-up [36].

The evolution of management strategies for CAA reflects growing understanding of natural history and risk factors. The shift from conservative management to earlier surgical intervention for high-risk subtypes parallels improved surgical outcomes and recognition of sudden death risk in seemingly stable patients. The integration of ECMO into management algorithms provides a safety net for acute decompensation, though prevention through timely diagnosis and intervention remains paramount.

This study has several important limitations that should be considered when interpreting our findings. The retrospective design introduces potential selection and information bias, though standardized data collection protocols minimize these effects. Because our institution

functions as a tertiary referral center, referral bias may have led to overrepresentation of more complex or high-risk cases, potentially limiting generalizability to broader pediatric populations. The single-center nature may limit generalizability, as referral patterns and management approaches vary among institutions. The relatively short follow-up period (median 2.3 years) precludes assessment of very long-term outcomes including late graft failure, exercise capacity in adulthood, and pregnancy-related complications. In addition, genetic testing was not systematically performed, which prevented evaluation of familial predisposition or syndromic contributions to coronary anomaly development. The small sample size for rare subtypes limits statistical power for subgroup analyses, particularly for other variants beyond the three major categories. Future multicenter studies with standardized genetic evaluation and extended follow-up are warranted to improve the generalizability and long-term predictive value of these findings. Additionally, the absence of systematic genetic testing prevents assessment of familial clustering or genetic modifiers of disease severity.

Future research directions emerging from our findings include the need for prospective multicenter registries to better define natural history and optimize risk stratification. Advanced imaging techniques including 4D flow MRI and computational modeling may provide personalized hemodynamic assessment to guide management decisions. The role of medical therapy in preventing progression and improving outcomes requires systematic investigation. Long-term neurodevelopmental outcomes following neonatal presentation and ECMO support warrant dedicated study. Finally, development of evidence-based guidelines specific to pediatric populations, incorporating age-specific considerations and growth potential, remains a critical need.

Conclusion

This comprehensive analysis of pediatric coronary artery anomalies demonstrates the heterogeneous nature of these conditions, with distinct patterns of presentation, risk profiles, and outcomes among anatomical subtypes. Early surgical intervention for high-risk variants, particularly ALCAPA and ALCA-RS with unfavorable anatomical features, results in excellent survival and functional outcomes in the modern era. ECMO provides crucial support for acute decompensation, though prevention through timely diagnosis and intervention remains optimal. Risk stratification incorporating anatomical, functional, and clinical parameters enables individualized management strategies that optimize outcomes while avoiding unnecessary intervention in low-risk patients. Continued long-term surveillance is essential for all patients, with particular attention to those with residual

ventricular dysfunction or high-risk anatomical features. These findings support the development of standardized protocols for diagnosis, risk assessment, and management of pediatric patients with coronary artery anomalies.

Acknowledgements

Not applicable.

Authors' contributions

Conception and design of the research: Li JC, Li SJ, Liu WQ. Acquisition of data: Liu WQ, Zhou JS, Xu YP, Qian XL. Analysis and interpretation of the data: Wang TJ, Meng FW. Statistical analysis: Liu WQ, Wang TJ, Qian XL. Obtaining financing: None. Writing of the manuscript: Liu WQ, Zhou JS. Critical revision of the manuscript for intellectual content: Li JC, Li SJ.

Funding

1. Henan Province Science and Technology Research Project: Application of ECMO Integrated Installation and Shipping Technology in Emergency and Critical Care Patients: No. LHGJ20220112.
2. Science and Technology Research Project of Henan Provincial Department of Science and Technology, "Application of Extracorporeal Membrane Oxygenation (ECMO) in Clinical Single Lung Transplantation" No. 182102310500.

Data availability

The datasets used and analyzed during the current study are available from the corresponding author on reasonable request.

Declarations

Ethics approval and consent to participate

This study was approved by the institutional ethics review board (approval number: [2024-24]). The requirement for written informed consent was waived because the study involved retrospective review of existing medical records with all data de-identified prior to analysis. All research activities complied with the ethical standards of the institutional and national research committees and with the 1964 Helsinki declaration and its later amendments.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

Received: 11 September 2025 / Accepted: 23 January 2026

Published online: 02 March 2026

References

1. Brothers JA, Frommelt MA, Jaquiss RDB, et al. Major coronary artery anomalies in a pediatric population: incidence and clinical importance detected by transthoracic echocardiography. *J Am Coll Cardiol*. 2001;37:593–7.
2. Brothers JA. Coronary artery anomalies in children: what is the risk? *Curr Opin Pediatr*. 2016;28:590–6.
3. Tang W, Xiang K, Zhou H, et al. Case report: surgical management of a rare right coronary artery fistula with a giant pseudoaneurysm compressing the pulmonary vein in a young patient. *Front Cardiovasc Med*. 2023;10:1233873.
4. Wu X, Qin K, Iroegbu CD, et al. Genetic analysis of potential biomarkers and therapeutic targets in ferroptosis from coronary artery disease. *J Cell Mol Med*. 2022;26(8):2177–90.
5. Tang S, Tang M, Daniel Iroegbu C, Yang J, Fan C. Case report: congenital coronary artery ring with single left coronary ostium and fistula: A previously unreported anatomy. *Front Cardiovasc Med*. 2021;8:699529.
6. Song L, Iroegbu CD, Fan C, Zhou X. Surgical management of a giant left sinus of Valsalva aneurysm with coronary artery abnormality. *Heart Surg Forum*. 2021;24(2):E296–8. Published 2021 Mar 25.

7. Wu S, Fan C, Yang J. A rare, giant coronary artery ectasia coexisting with a coronary artery fistula in an older infant. *Cardiol Young*. 2017;27(7):1387–9.
8. Molossi S, Agrawal H, Mery CM, et al. Outcomes in anomalous aortic origin of a coronary artery following a prospective standardized approach. *Circ Cardiovasc Interv*. 2020;13:e008445.
9. Doan TT, Zea-Vera R, Agrawal H, et al. Ischemia in anomalous aortic origin of a right coronary artery: large pediatric cohort medium-term outcomes. *Circ Cardiovasc Interv*. 2023;13:e008375.
10. Jegatheeswaran A, Devlin PJ, Williams WG, et al. Outcomes after anomalous aortic origin of a coronary artery repair: a congenital heart surgeons' society study. *J Thorac Cardiovasc Surg*. 2020;160:757–71.
11. Gräni C, Padalino MA. Editorial: coronary artery anomalies: a 2020 review. *Front Cardiovasc Med*. 2022;9:776951.
12. Borns J, Gräni C, Kadner A, et al. Symptomatic coronary anomalies and ischemia in teenagers – rare but real. *Front Cardiovasc Med*. 2020;7:559794.
13. Harky A, Noshirwani A, Karadakhly O, Ang J. Comprehensive literature review of anomalies of the coronary arteries. *J Card Surg*. 2019;34(11):1328–43.
14. Mery CM, De León LE, Molossi S, et al. Outcomes of surgical intervention for anomalous aortic origin of a coronary artery: a large contemporary prospective cohort study. *J Thorac Cardiovasc Surg*. 2018;155:305–19.
15. Hatoum H, Krishnamurthy R, Parthasarathy J, et al. Flow dynamics in anomalous aortic origin of a coronary artery in children: importance of the intramural segment. *Semin Thorac Cardiovasc Surg*. 2022;34(1):226–35.
16. Bigler MR, Ashraf A, Seiler C, et al. Hemodynamic relevance of anomalous coronary arteries originating from the opposite sinus of Valsalva-in search of the evidence. *Front Cardiovasc Med*. 2021;7:591326.
17. Blickenstaff EA, Smith SD, Cetta F, Connolly HM, Majdalany DS. Anomalous left coronary artery from the pulmonary artery: how to diagnose and treat. *J Pers Med*. 2023;13(11):1561.
18. Luo H, Kwaku OE, Lai Y, Yue R. Adult-type anomalous origin of the left coronary artery from the pulmonary artery and right coronary-right atrial fistula: a case report. *BMC Cardiovasc Disord*. 2024;24(1):31.
19. Fan C, Tang M, Yang J. Right coronary artery fistula into left ventricle: blood flow steal phenomenon. *Ann Thorac Surg*. 2017;103(3):e287.
20. Brothers JA, Gaynor JW, Paridon S, et al. Expert consensus guidelines: anomalous aortic origin of a coronary artery. *J Thorac Cardiovasc Surg*. 2017;153:1440–57.
21. Patlolla SH, Stephens EH, Schaff HV, et al. Outcomes of a protocolized approach for surgical unroofing of intramural anomalous aortic origin of coronary artery in children and adults. *J Thorac Cardiovasc Surg*. 2023;165(5):1641–50.
22. Roeleveld PP, Mendonca M. Neonatal cardiac ECMO in 2019 and beyond. *Front Pediatr*. 2019;7:327.
23. Jacobs JP, Franklin RC, Jacobs ML, et al. Classification of the functionally univentricular heart: unity from mapped codes. *Cardiol Young*. 2006;16(Suppl 1):9–21.
24. Jinmei Z, Yunfei L, Yue W, Yongjun Q. Anomalous origin of the left coronary artery from the pulmonary artery (ALCAPA) diagnosed in children and adolescents. *J Cardiothorac Surg*. 2020;15(1):90.
25. Naimo PS, Fricke TA, d'Udekem Y, et al. Surgical intervention for anomalous origin of left coronary artery from the pulmonary artery in children: a long-term follow-up. *Ann Thorac Surg*. 2016;101:1842–8.
26. Doan TT, Wilkes JK, Reaves O'Neal DL, et al. Clinical presentation and Medium-Term outcomes of children with anomalous aortic origin of the left coronary artery: High-Risk features beyond interarterial course. *Circ Cardiovasc Interv*. 2023;16(5):e012635.
27. Bigler MR, Kadner A, Räber L, et al. Therapeutic management of anomalous coronary arteries originating from the opposite sinus of valsalva: current Evidence, proposed Approach, and the unknowing. *J Am Heart Assoc*. 2022;11(20):e027098.
28. Doan TT, Puelz C, Rusin C, Molossi S. Anomalous aortic origin of a coronary artery in pediatric patients. *Curr Pediatr Rep*. 2024;12:69–80.
29. Evangelista M, Ferrero P, D'Aiello AF, et al. Coronary artery anomalies: what are they? When to suspect? How to treat?-a narrative review. *Transl Pediatr*. 2024;13:1242–57.
30. Bonilla-Ramirez C, Molossi S, Sachdeva S, et al. Outcomes in anomalous aortic origin of a coronary artery after surgical reimplantation. *J Thorac Cardiovasc Surg*. 2021;162:1191–9.
31. Bibevski S, Ruzmetov M, Turner II, Scholl FG. Anomalous aortic origin of right coronary artery: outcomes of surgical and nonsurgical treatment. *Ann Thorac Surg*. 2022;114(6):2338–45.
32. Amodeo A, Stojanovic M, Erdil T, et al. Risk factors and outcomes of children with congenital heart disease on extracorporeal membrane Oxygenation-A Ten-Year Single-Center report. *Life (Basel)*. 2023;13(7):1582.
33. Ponzoni M, Frigo AC, Padalino MA. Surgery for anomalous aortic origin of a coronary artery (AAOCA) in children and adolescents: a meta-analysis. *World J Pediatr Congenit Heart Surg*. 2022;13:485–94.
34. Padalino MA, Franchetti N, Hazekamp M, et al. Surgery for anomalous aortic origin of coronary arteries: a multicentre study from the European congenital heart surgeons Association†. *Eur J Cardiothorac Surg*. 2019;56(4):696–703.
35. Law T, Dunne B, Stamp N, et al. Surgical results and outcomes after reimplantation for the management of anomalous aortic origin of the right coronary artery. *Ann Thorac Surg*. 2016;102:192–8.
36. Herrmann JL, Goldberg LA, Khan AM, et al. A comparison of perioperative management of anomalous aortic origin of a coronary artery between an adult and pediatric cardiac center. *World J Pediatr Congenit Heart Surg*. 2016;7:721–6.

Publisher's Note

Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.